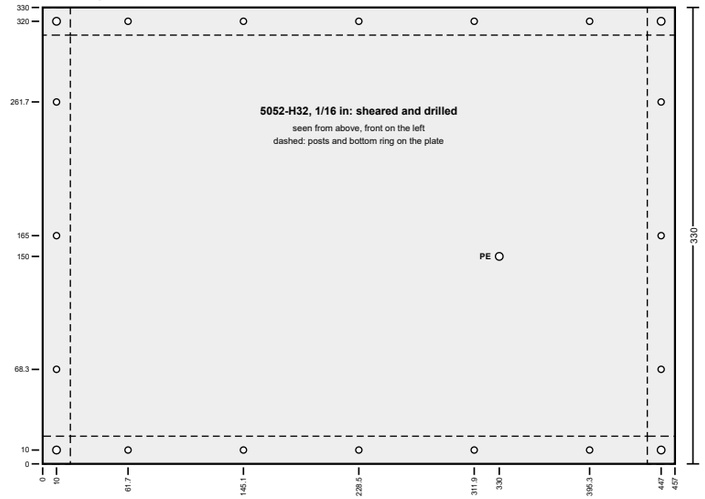
# RMR control panel: frame and deck plates

2020 frame 457 x 330 x 279.4 mm (inside 18 x 13 x 11 in) in two modules, on two 1/16 in 5052 deck plates, with 1/8 in acrylic walls. Views 1:5, detail E 2:1, printed at 100 %. Millimetres.

## A Base plate, lower deck

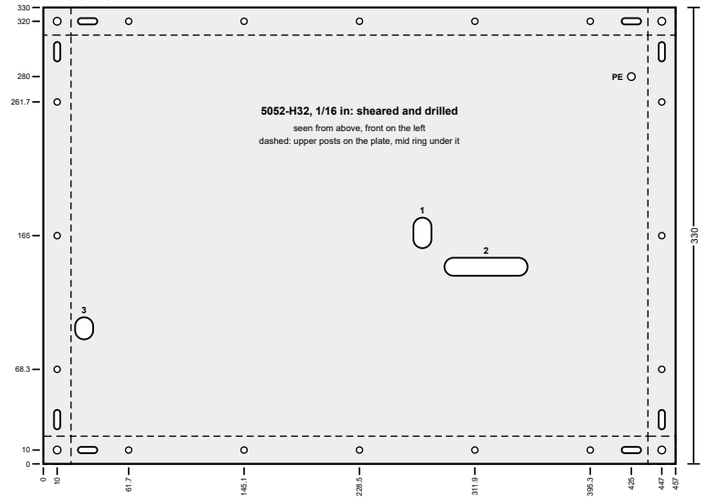
deck_base_plate.dxf




4 x Ø5.5: M5 into the lower posts, with the feet
16 x Ø4.5: M4 x 6 into the bottom ring T-nuts, from below
Ø5.5 at (330, 150): PE stud, first screw of the lower ground bar
Drill through the acrylic template (sheet B's DXF plus this PE stud).
DIN rail and duct holes from the final layout: M4 x 12, nyloc nut below; the tails end 2.6 mm or more over the floor with the plate's sag (NutFits)
Ground bar screws, both plates: M5, plain nut below, tooth washer under the bar; the bar's foot + 8.5 to + 13.2 long (OpenScrews)

## B Upper plate

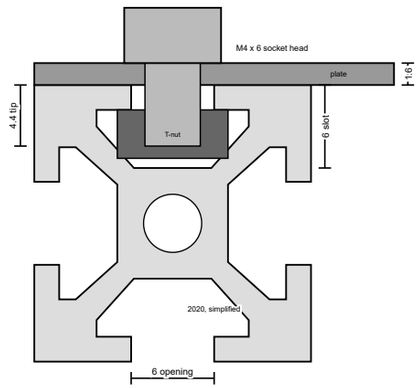
deck_upper_plate.dxf




4 x Ø5.5: the locating studs. 16 x Ø4.5: M4 x 6 into the mid ring T-nuts
8 slots 4.5 wide, 25 to 39 from the edges: upper-post bracket screws, M4 x 10. PE stud at (425, 280)
Template: cut this DXF from 1/8 in acrylic; drill, then nibble the slots to it.
DIN rail and duct holes from the final layout: M4 x 12, nyloc below, 15 mm or more from each end
of the run; the tails end 20 mm or more over what stands below (HangGaps)
Shield clamp: M4 into a nyloc, the clamp's base + 8 to + 61.6 long; ground bar screws off the mid ring and its corner brackets (OpenScrews)
Cable slots, round ends, edge trim: 1: x 267.5-280.5, y 156-178; 2: x 290-350, y 136-149; 3: x 23-36, y 90-106
1: driver 24 V feeds, 0V-P returns. 2: SSR feeds and returns, servo 6 V. 3: X2, the E-stop latch, pin 43 sense. None over the 110 VAC compartment (y 182 up).

## E Plate on a ring rail

2:1




Euro 2020, 6 mm slot. Depth 6, lip 1.8, bore 4.2:
assumed; measure the first stick.
Base plate: the same joint upside down, from below.

### Two modules

Like a platform-framed house. Lower module: base plate, bottom ring, 4 lower posts. Upper module, built and wired on the bench: mid ring (the platform), upper plate, 4 upper posts standing on the plate, lid ring. It drops onto 4 M5 studs (50 long) in the lower post tops; each passes the mid ring and the plate into a Ø5.5 socket in its upper post, so the decks line up within about 0.3 mm. 8 brackets from the lower posts to the mid ring hold it down.

### Joints

Corner brackets on the kit's M5 screws and sliding T-nuts, slid in before a joint closes: 8 on the bottom ring, 8 lower posts to mid ring, 4 in the mid ring, 8 upper posts (rail legs on the plate), 4 in the lid ring. Each plate, screwed every 97 mm or less, keeps its ring square: no diagonal braces.

### Screws, checked by ScrewFits()

Each stops 0.5 short of the slot floor and catches 3 or more threads of its spring T-nut:
base plate: M4 x 6, tip 4.4
upper plate: M4 x 6, tip 4.4
bracket leg: M4 x 10, tip 4.9
acrylic panel: M5 x 8, tip 4.8
back panel + washer: M5 x 8, tip 5.4
mount, front panel: M5 x 12, tip 5.3
mount, back panel: M5 x 10, tip 4.9
mount, gantry: M5 x 8, tip 4.5
Lid M5 x 35 needs 15 of end tap; feet M5 x 16.

### PE bonding

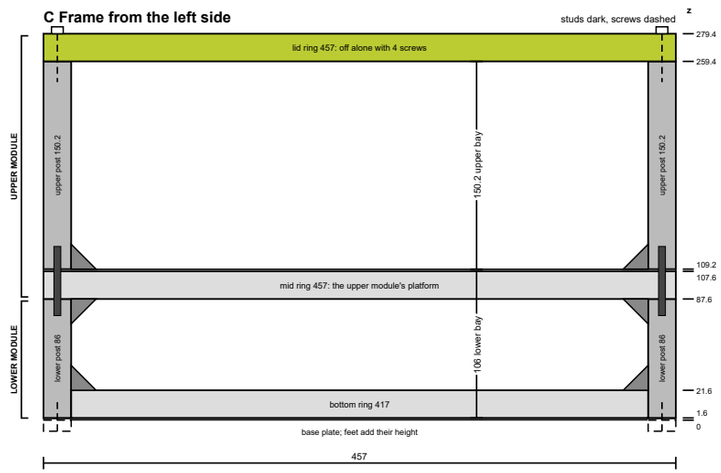
Anodize insulates: each rail is its own part. Drill Ø4.2 through a slot floor into the bore, tap M5; M5 x 16 with a ring lug and a tooth washer. One 12 AWG home run per part to its module's bar: lower 17 parts, 2 bars; upper 26, 2 bars. DIN rails through their PE terminals, TB6600s under an ear screw. Measure each part to PE at 0.2 to 10 A (IEC 60204-1 18.2.2).

### Stiffness

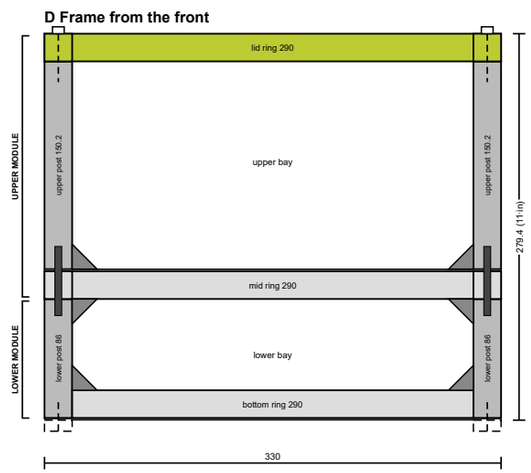
5 kg on a plate, simply supported on its ring (Timoshenko Table 8): 0.9 mm for 1/16 in 5052.

## C Frame from the left side

studs dark, screws dashed




## D Frame from the front




### Cut list: 2020, 6 mm slot, from 1000 mm sticks

| Part | Qty | Length | Ends |
|---|---|---|---|
| Lower post | 4 | 86 | tap M5 both ends |
| Upper post | 4 | 150.2 | tap M5 top; drill Ø5.5 x 19 bottom |
| Bottom ring, long | 2 | 417 | between the lower posts |
| Ring rail, short | 6 | 290 | bottom, mid and lid ring |
| Mid and lid ring, long | 4 | 457 | drill Ø5.5 through, 10 from ends |

5347 mm in 20 pieces from 6 sticks (kerf 3, one squaring cut each):
1: 457 + 457; 2: 457 + 457; 3: 417 + 417 + 150.2;
4: 290 + 290 + 290 + 86; 5: 290 + 290 + 290 + 86; 6: 150.2 + 150.2 + 150.2 + 86 + 86.
Cut equal parts against one stop: the plates set the squareness.

### Plates and panels

| Part | Nest sheet | Size | Material |
|---|---|---|---|
| Base plate | aluminum 1 | 457 x 330 | 5052, 1/16 in |
| Upper plate | aluminum 2 | 457 x 330 | 5052, 1/16 in |
| Lower side, left | acrylic 1 | 457 x 87.6 | acrylic, 1/8 in |
| Lower side, right | acrylic 3 | 457 x 87.6 | acrylic, 1/8 in |
| Lower front | acrylic 1 | 330 x 87.6 | acrylic, 1/8 in |
| Lower back | aluminum 1 | 330 x 87.6 | 5052, 1/16 in |
| Upper side, left | acrylic 2 | 457 x 191.8 | acrylic, 1/8 in |
| Upper side, right | acrylic 2 | 457 x 191.8 | acrylic, 1/8 in |
| Upper front | acrylic 3 | 330 x 191.8 | acrylic, 1/8 in |
| Upper back | acrylic 3 | 330 x 191.8 | acrylic, 1/8 in |
| Top, on the lid ring | acrylic 1 | 457 x 330 | acrylic, 1/8 in |
| Grommet ring x 5 | acrylic 2 | Ø64, bore 49.8 | acrylic, 1/8 in |

Acrylic panels on 3 sheets of 24 x 18 in, each inside the 813 x 457 laser bed: one DXF per sheet, panels_acrylic_1.dxf to _3.dxf. The aluminum lower back panel carries the inlet and the door cable's M20 gland (every mains part sits on metal); drill and nibble it through panel_lower_back_template.dxf, cut from the spare sheet. Each part is drawn from outside: u along x (front at 0) or y (left at 0), v up; turn it over to match. Screws Ø5.5 on the slot centre lines, 10 from each post end, ring screws at an even pitch, except the lower front and back panels: their bottom post screws sit at 30 in vertical slots, through the gantry mount brackets, and they have no ring screw (sheet 5); the top has 8 into the lid ring and Ø10 over each lid screw. Vents: 21 slots 4 x 88 on 8 over the TB6600s. Cable entries: 5 desk grommets in Ø50 holes (two upper left, upper front, upper right, lower left), each held by a Ø64 acrylic ring pressed onto its sleeve behind the wall; caliper one grommet before cutting. Q0 is the inlet's switch if that is 2-pole. Read back from the DXFs: least web 4 mm (2 needed). Order list: bom_frame.csv. NOT CUT YET, placeholders until the owner's data: the PSU fan vent (its diameter), the IEC inlet with switch cut-out. Hold the lower side, right (acrylic 3) and the back panel template until then.

### Assembly

1. Caliper a grommet and K2 before cutting: the lower grommet's ring sets the lower posts (86). Measure the first stick and a bracket leg, then rerun ScrewFits().
2. Cut the list, equal parts against one stop. Tap and drill the ends; drill and tap one bond hole in every rail.
3. Lower module: feet and lower posts on the base plate, bottom ring on brackets, plate screws from below, studs in with threadlocker, ground bar on the PE stud.
4. Build the lower deck, every 110 VAC terminal finger-safe or covered (ADR-0013); bond and measure it.
5. Upper module, on the bench: mid ring on its corner brackets, upper plate on it, upper posts over the stud holes (a loose stud as a pin) with their brackets, then the lid ring. Ground bars on the upper PE stud; wire it.
6. Set it on the studs; 8 brackets to the lower posts; plug the inter-deck terminals; clamp the PE link; measure; panels last.

## Frame and deck plates

Sheet 4. Legal, landscape, 100 %.
Views 1:5, detail E 2:1. Redrawn 2026-10-19.
Two modules. Design, not yet built.
DXFs: both deck plates, panel sheets, template.